

Sprint 5, MongoDB.

Nivel 1:

Crea una base de datos con MongoDB utilizando como colecciones los archivos adjuntos.

Llamamos a la BBDD creada CinemaDB y cargamos los archivos correspondientes:

ConnectionsEditViewHelp

localhost:27017CinemaDB

moviescommentssessionsuserstheaters

Compass

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

localhost:27017

CinemaDB

comments

movies

sessions

theaters

users

admin

config

local

localhost:27017CinemaDB

Open MongoDB shellCreate collectionRefresh

Collection name	Properties	Storage size	Data size	Documents	Avg. document size	Indexes	Total index size
comments	-	4.10 kB	14.29 MB	50K	284.00 B	1	4.10 kB
movies	-	4.10 kB	37.62 MB	24K	1.60 kB	1	4.10 kB
sessions	-	4.10 kB	540.00 B	1	540.00 B	1	4.10 kB
theaters	-	4.10 kB	349.83 kB	1.6K	223.00 B	1	4.10 kB
users	-	4.10 kB	29.57 kB	185	159.00 B	1	4.10 kB

Ejercicio 1.1 Muestra los 2 primeros comentarios que hay en la base de datos.

Seleccionamos la colección 'comments', podemos realizar el ejercicio desde Documents -> Limit 2 o desde Aggregations y filtrando desde Stage1. Visualizamos los 2 resultados.

The screenshot shows the MongoDB Compass interface for the 'comments' collection in the 'CinemaDB' database on 'localhost:27017'. The 'Aggregations' tab is active, and a pipeline with one stage, '\$limit', is defined. The output preview shows two documents:

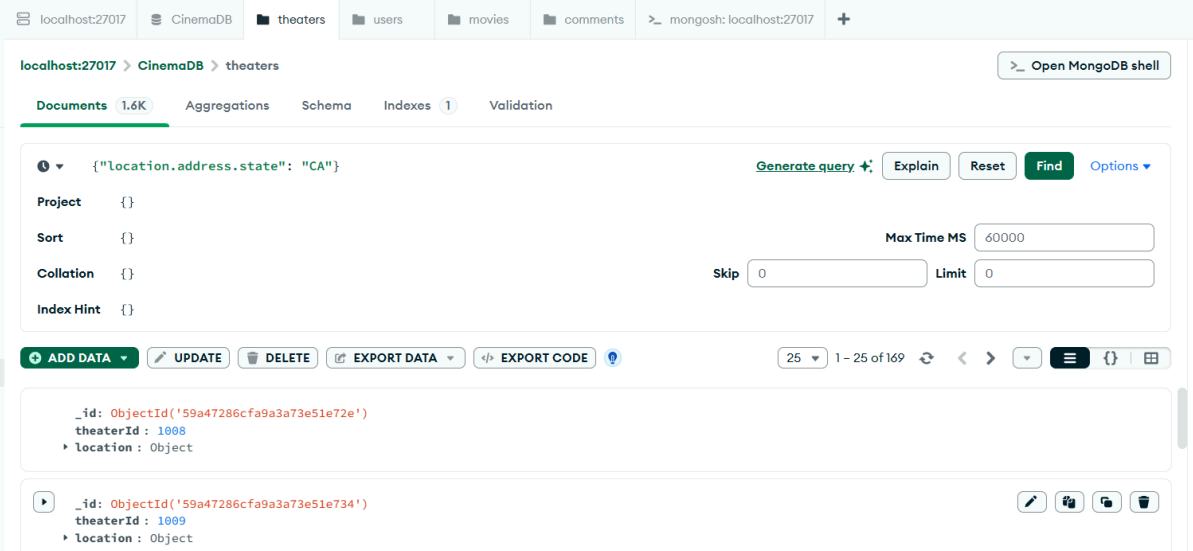
```
{ "_id": "ObjectId('5a9427648b0beebe69579cc')", "name": "Andrea Le", "email": "andrea_le@fakegmail.com", "movie_id": "ObjectId('573a139ef29313caabcd418...", "text": "Rem officiiis eaque repellendus amet eos doloribus. Porro dolor volupta...", "date": "2012-03-26T23:20:16.000+00:00" }, { "_id": "ObjectId('5a9427648b0beebe69579cf...", "name": "Greg Powell", "email": "greg_powell@fakegmail.com", "movie_id": "ObjectId('573a139ef29313caab...", "text": "Tenetur dorum molestiae ea. Eligendi praesentium unde quod p Co...", "date": "1987-02-10T00:29:36.000+00:00" }
```

1.2 ¿Cuántos usuarios tenemos registrados?

Desde colección users, pestaña Documents (no filtros) -> find, visualizamos 185 usuarios

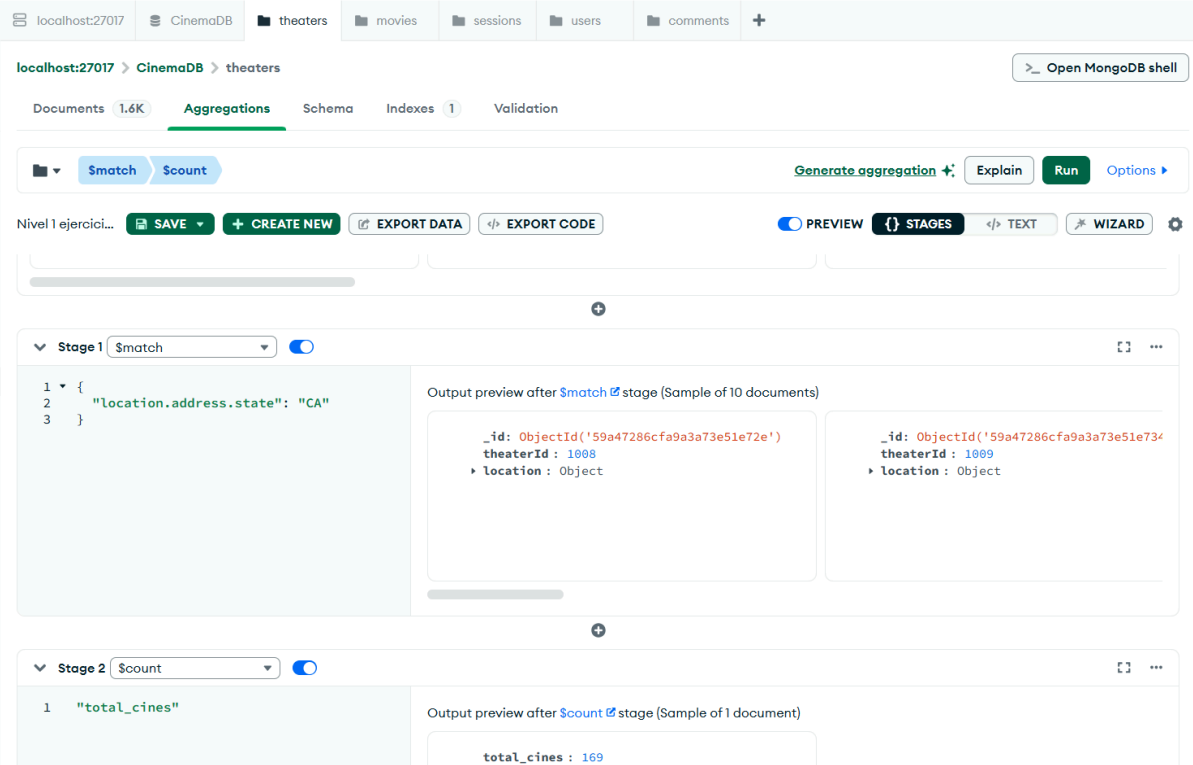
The screenshot shows the MongoDB Compass interface for the 'users' collection in the 'CinemaDB' database on 'localhost:27017'. The 'Documents' tab is active, and it shows 185 documents. The breadcrumb navigation at the top reads 'localhost:27017 > CinemaDB > users'.

1.3 ¿Cuántos cines hay en el estado de California?



The screenshot shows the MongoDB Compass interface. The top navigation bar includes 'localhost:27017', 'CinemaDB', and a list of collections: 'theaters', 'users', 'movies', 'comments', and 'mongodb:localhost:27017'. The 'theaters' collection is selected. The 'Documents' tab is active, showing a list of documents. A query filter is applied: `{ "location.address.state": "CA" }`. The 'Find' button is highlighted. Below the query, there are fields for 'Project', 'Sort', 'Collation', and 'Index Hint', all set to {}. On the right, there are fields for 'Max Time MS' (60000), 'Skip' (0), and 'Limit' (0). At the bottom, there are buttons for 'ADD DATA', 'UPDATE', 'DELETE', 'EXPORT DATA', and 'EXPORT CODE'. The document list shows two documents with their IDs, theater IDs, and location objects.

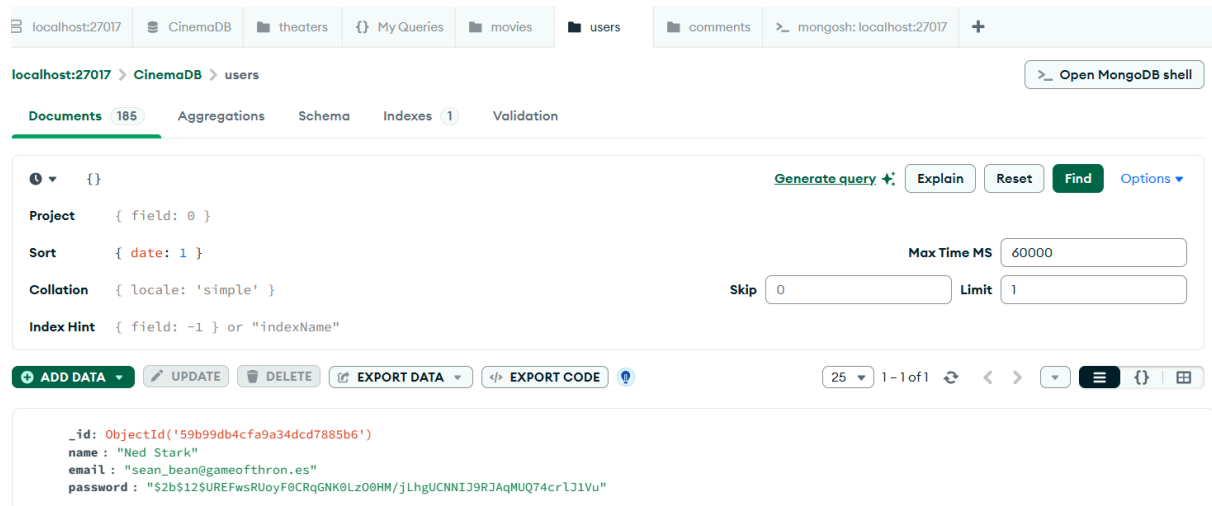
Dentro de la colección 'theaters' desde Documents con filtro o desde Aggregations con \$match filtramos y \$count contabilizamos podemos visualizar resultado: 169



The screenshot shows the MongoDB Compass interface with the 'Aggregations' tab selected. The 'theaters' collection is selected. The aggregation pipeline is defined with two stages: Stage 1 is '\$match' with a filter of `{ "location.address.state": "CA" }`, and Stage 2 is '\$count'. The 'Run' button is highlighted. Below the aggregation pipeline, there are buttons for 'SAVE', 'CREATE NEW', 'EXPORT DATA', and 'EXPORT CODE'. The 'PREVIEW' button is also highlighted. The output preview for Stage 1 shows a sample of 10 documents. The output preview for Stage 2 shows a single document with the field `total_cines` and the value 169.

1.4 ¿Cuál fue el primer usuario/a en registrarse?

Desde Documents filtramos fecha y limitamos a 1,



localhost:27017 > CinemaDB > users

Documents 185 Aggregations Schema Indexes 1 Validation

Generate query + Explain Reset Find Options ▾

Project { field: 0 }

Sort { date: 1 }

Collation { locale: 'simple' }

Index Hint { field: -1 } or "indexName"

Max Time MS 60000

Skip 0 Limit 1

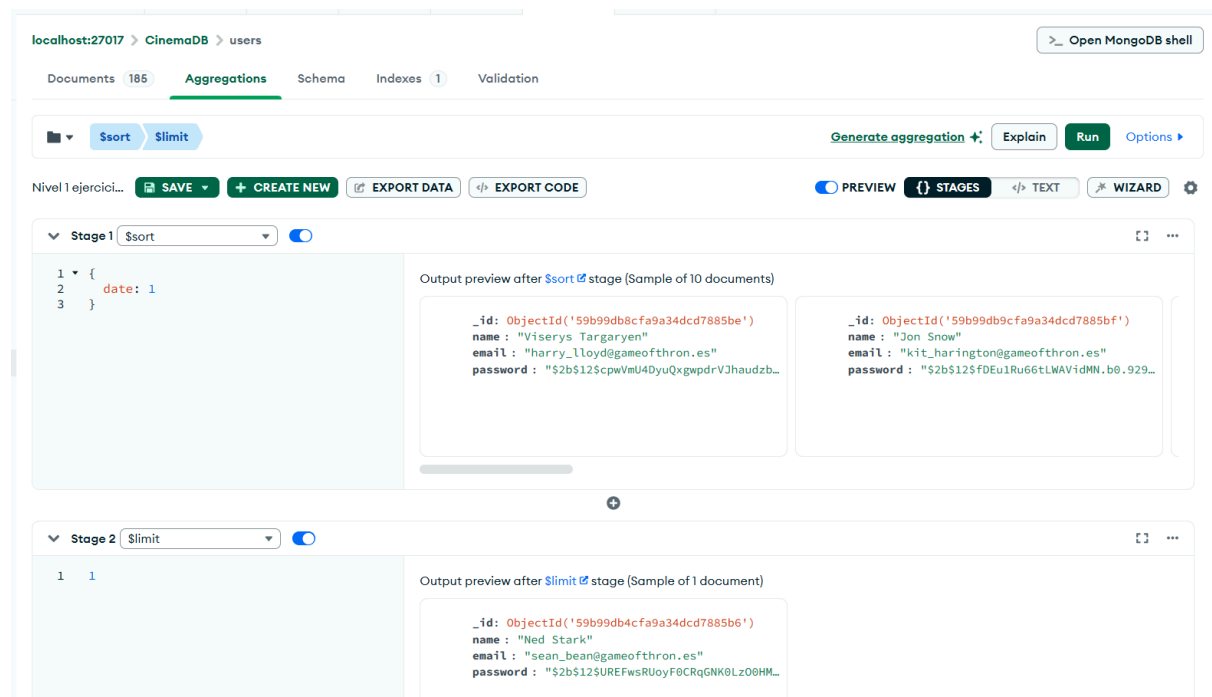
ADD DATA UPDATE DELETE EXPORT DATA EXPORT CODE

25 1 - 1 of 1

```
{
  "_id": ObjectId('59b99db4cfa9a34dcd7885b6'),
  "name": "Ned Stark",
  "email": "sean_bean@gameofthron.es",
  "password": "$2b$12$SUREFwsRUoyF0CRqGNK0Lz00HM/jLhgUCNNI39RJAqMUQ74crLJ1Vu"
```

otra opción es con sort y limit desde Aggregations: 'Ned Stark'

+



localhost:27017 > CinemaDB > users

Documents 185 Aggregations Schema Indexes 1 Validation

Generate aggregation + Explain Run Options ▾

Nivel 1 ejercicio... SAVE CREATE NEW EXPORT DATA EXPORT CODE

PREVIEW STAGES TEXT WIZARD

Stage 1 \$sort

1 { date: 1 }
2 { }
3 { }

Output preview after \$sort stage (Sample of 10 documents)

```
{
  "_id": ObjectId('59b99db8cfa9a34dcd7885be'),
  "name": "Viserys Targaryen",
  "email": "harry_lloyd@gameofthron.es",
  "password": "$2b$12$cpwVmU4DyuQxgwpdrVJhaudzb..."
}
```

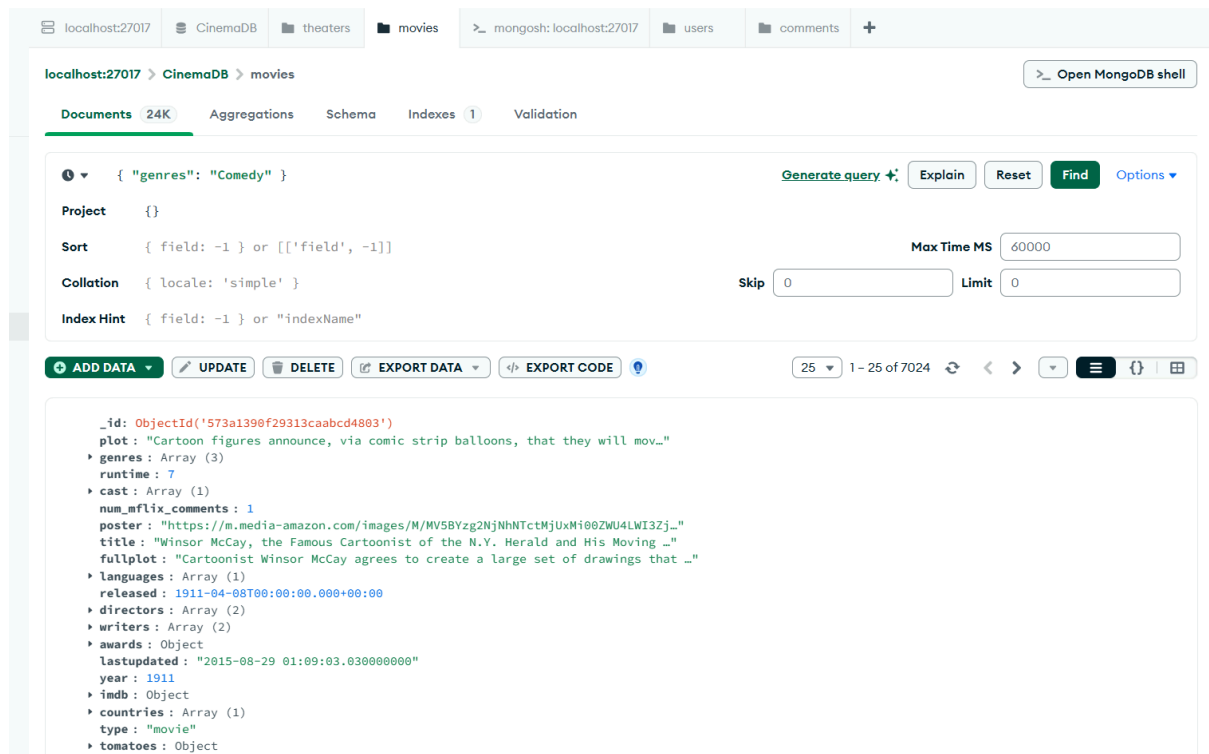
Stage 2 \$limit

1 1

Output preview after \$limit stage (Sample of 1 document)

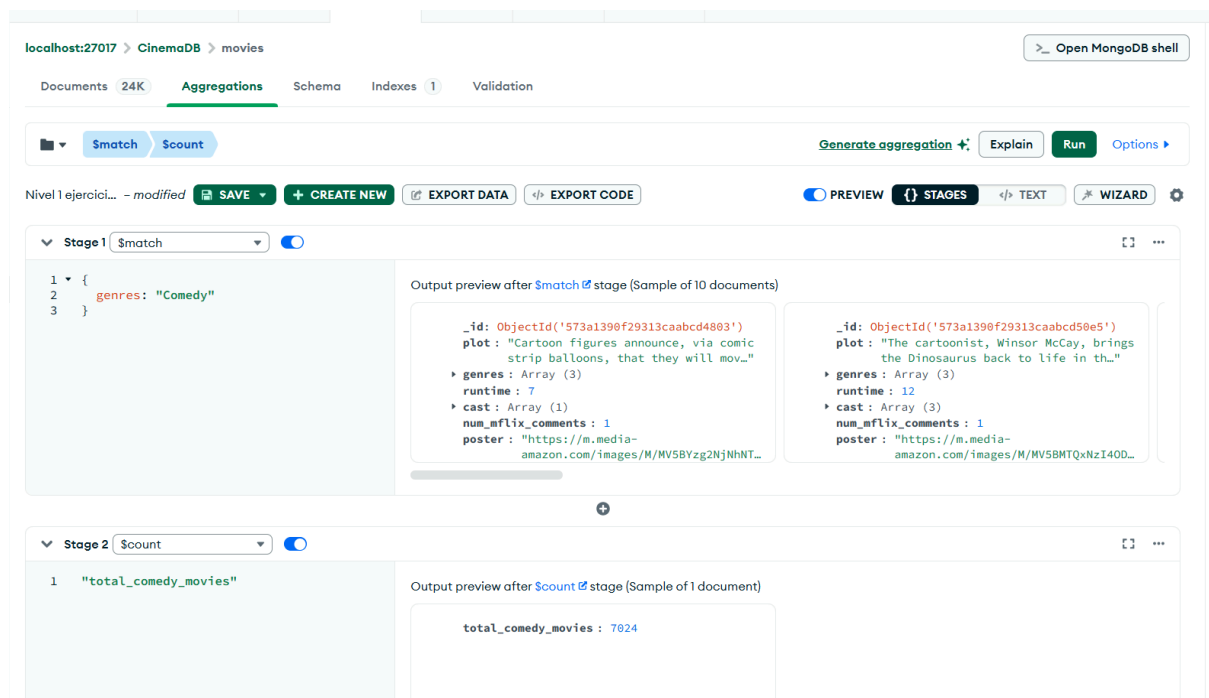
```
{
  "_id": ObjectId('59b99db4cfa9a34dcd7885b6'),
  "name": "Ned Stark",
  "email": "sean_bean@gameofthron.es",
  "password": "$2b$12$SUREFwsRUoyF0CRqGNK0Lz00HM..."
}
```

1.5 ¿Cuántas películas de comedia hay en nuestra base de datos?



The screenshot shows the MongoDB Compass interface. The top navigation bar includes 'localhost:27017', 'CinemaDB', and a list of collections: 'theaters', 'movies', 'users', and 'comments'. The 'movies' collection is selected, showing 24K documents. The query tab is active, displaying a query: `{ "genres": "Comedy" }`. The 'Project' field is empty, 'Sort' is set to `{ field: -1 } or [['field', -1]]`, 'Collation' is `{ locale: 'simple' }`, and 'Index Hint' is `{ field: -1 } or "indexName"`. The 'Max Time MS' is set to 60000. The 'Skip' and 'Limit' fields are both 0. Below the query, there are buttons for 'ADD DATA', 'UPDATE', 'DELETE', 'EXPORT DATA', and 'EXPORT CODE'. The results section shows a sample document for a movie with the following fields: `_id`, `plot`, `genres`, `runtime`, `cast`, `num_mflix_comments`, `poster`, `title`, `fullplot`, `languages`, `released`, `directors`, `writers`, `awards`, `lastupdated`, `year`, `imdb`, `countries`, `type`, and `tomatoes`.

Agregación en dos etapas, aplicamos `$match` para filtrar género comedia y `$count` para contabilizar la condición, resultado 7024



The screenshot shows the MongoDB Compass interface with the 'Aggregations' tab selected. The pipeline consists of two stages: `$match` and `$count`. The `$match` stage has a query: `{ "genres": "Comedy" }`. The `$count` stage has a query: `"total_comedy_movies"`. The 'Output preview after \$match stage (Sample of 10 documents)' shows two sample documents. The 'Output preview after \$count stage (Sample of 1 document)' shows the result: `total_comedy_movies : 7024`.

Ejercicio 2

Muéstrame todos los documentos de las películas producidas en 1932, pero que el género sea drama o estén en francés.

Primero filtramos por año y después por las condiciones:

The screenshot shows the MongoDB Compass interface. The 'movies' collection is selected. A query is entered in the 'Find' tab: `{ "year": 1932, "$or": [ { "genres": "Drama" }, { "languages": "French" } ] }`. The 'Project' field is empty, 'Sort' is set to `{ field: -1 } or [['field', -1]]`, and 'Collation' is `{ locale: 'simple' }`. The 'Index Hint' is `{ field: -1 } or "indexName"`. The 'Max Time MS' is 60000, 'Skip' is 0, and 'Limit' is 0. The 'Find' button is highlighted. Below the query, a sample document is shown:

```
{
  "_id": ObjectId('573a1391f29313caabcd9458'),
  "plot": "A young artist draws a face at a canvas on his easel. Suddenly the mou...",
  "runtime": 55,
  "rated": "UNRATED",
  "cast": Array (4),
  "num_mflix_comments": 1,
  "poster": "https://m.media-amazon.com/images/M/MV5BYWY3ODE5ZWYtYjlmYi00NjA4LTk4ZW...",
  "title": "The Blood of a Poet",
  "lastupdated": "2015-09-16 13:13:05.537000000",
  "languages": Array (1),
  "released": 2010-05-20T00:00:00.000+00:00,
  "directors": Array (1),
  "writers": Array (1),
  "awards": Object,
  "year": 1932
}
```

desde Aggregations \$match filtra el año y las la primera condición o la segunda, \$project muestra los campos seleccionados

The screenshot shows the MongoDB Compass interface with the 'Aggregations' tab selected. The pipeline consists of two stages: `$match` and `$project`. The `$match` stage has the same query as in the first screenshot. The `$project` stage is defined as `{ title: 1, year: 1, genres: 1, languages: 1 }`. The 'Run' button is highlighted. Below the pipeline, the output preview for each stage is shown. The output of the `$match` stage shows a sample of 10 documents, including the first document from the first screenshot. The output of the `$project` stage shows a sample of 10 documents, including the first document from the first screenshot, with only the specified fields (title, year, genres, languages) projected.

este es el 'run' de la pipeline:

The screenshot shows the MongoDB Compass interface for the 'CinemaDB' database, 'movies' collection. The 'Aggregations' tab is active, showing a pipeline with one stage. The 'Run' button is highlighted in green. The interface includes a top bar with 'localhost:27017 > CinemaDB > movies', a top navigation bar with 'Documents 24K', 'Aggregations', 'Schema', 'Indexes 1', and 'Validation'. Below the navigation bar are buttons for 'Smatch', 'Sproject', and 'Edit'. The main area shows the aggregation pipeline with one stage. The 'Run' button is highlighted in green. The bottom bar shows 'ALL RESULTS', 'EXPORT DATA', and 'EXPORT CODE' buttons. The status bar indicates 'Showing 1 - 18 count results'.

Ejercicio 3

Muéstrame todos los documentos de películas estadounidenses que tengan entre 5 y 9 premios que fueron producidas entre 2012 y 2014.

Desde Documents de la colección 'movies', movies filtramos el país, 'USA', el número de premios ganados (entre 5 y 9) y el año de publicación (debe de ser entre 2012-2014). Resultado 166 películas.

The screenshot shows the MongoDB Compass interface for the 'CinemaDB' database, 'movies' collection. The 'Documents' tab is active, showing a list of documents. A query filter is applied: `{ "countries": "USA", "awards.wins": { "$gte": 5, "$lte": 9 }, "year": { "$gte": 2012, "$lte": 2014 } }`. The 'Find' button is highlighted in green. The interface includes a top bar with 'localhost:27017 > CinemaDB > movies', a top navigation bar with 'Documents 24K', 'Aggregations', 'Schema', 'Indexes 1', and 'Validation'. Below the navigation bar are buttons for 'ADD DATA', 'UPDATE', 'DELETE', 'EXPORT DATA', and 'EXPORT CODE'. The main area shows the query filter and the 'Find' button. The bottom bar shows '25' documents, '1 - 25 of 166', and navigation buttons. The status bar indicates 'Showing 1 - 25 of 166'.

Nivel 2 Ejercicio 1: Cuenta cuántos comentarios escribe un usuario/a que utiliza "GAMEOFTHRON.ES" como dominio de correo electrónico.

The screenshot shows the MongoDB Compass interface. The database is 'CinemaDB' and the collection is 'comments'. A query is entered in the 'Find' tab: `{ email: RegExp("@gameofthron\\.es$", "i") }`. The query is executed, and the results are displayed in a table. The first two documents are shown:

```
{ "_id": ObjectId("5a9427648b0beebe69579d0"), "name": "Talisa Maegyr", "email": "oona_chaplin@gameofthron.es", "movie_id": ObjectId("573a1390f29313caabcd41b1"), "text": "Rem itaque ad sit rem voluptatibus. Ad fugiat maxime illum optio iure ...", "date": "1998-08-22T11:45:03.000+00:00" }
```

```
{ "_id": ObjectId("5a9427648b0beebe69579d5"), "name": "Petr Baelish", "email": "aidan_gillen@gameofthron.es", "movie_id": ObjectId("573a1390f29313caabcd4218"), "text": "Quo deserunt ipsum ipsum. Tenetur eos nemo nam sint praesentium minus ...", "date": "2001-07-13T19:25:09.000+00:00" }
```

En esta agregación desde la colección comments filtramos \$match para encontrar comentarios cuyo mail termina con el indicado y \$count contabiliza los que cumplen esa condición. Total comentarios 22.841

The screenshot shows the MongoDB Compass interface with an aggregation pipeline defined in the 'Aggregations' tab. The pipeline consists of two stages:

- Stage 1: \$match** with the query: `{ email: { $regex: '/@gameofthron\\.es$/i' } }`. The output preview shows a sample of 10 documents.
- Stage 2: \$count** with the expression: `"total_comments"`. The output preview shows a sample of 1 document: `{ "total_comments": 22841 }`.

Ejercicio 2

¿Cuántos cines hay en cada código postal situados dentro del estado Washington D. C. (DC)?

En el stage1 filtramos (\$match) el estado 'DC' y en stage2 agrupamos (\$group) por código postal

The screenshot shows the MongoDB Aggregations Builder interface. At the top, the path is 'localhost:27017 > CinemaDB > theaters'. The 'Aggregations' tab is selected, showing 1.6K documents. The aggregation pipeline consists of two stages:

- Stage 1 (\$match):** The query is `{ "location.address.state": "DC" }`. The output preview shows three documents with their IDs, theater IDs, and location objects.
- Stage 2 (\$group):** The query is `{ "_id": "$location.address.zipcode", "total_cines": { $sum: 1 } }`. The output preview shows three documents with their ZIP codes and the count of theaters.

Aplicamos el Run para ver el resultado:

The screenshot shows the MongoDB Aggregations Builder interface after clicking the 'Run' button. The 'Run' button is highlighted in green. Below the aggregation pipeline, the 'ALL RESULTS' section displays the output of the aggregation:

```

_id: "20002"
total_cines: 1

_id: "20010"
total_cines: 1

_id: "20016"
total_cines: 1

```

Nivel 3 Ejercicio 1

Encuentra todas las películas dirigidas por John Landis con una puntuación IMDb (Internet Movie Database) de entre 7,5 y 8.

Seleccionamos las películas del campo directors con valor "John Landis", después filtramos por puntuación y añadimos el baremo. Desde Aggregations -> Stage1 -> \$match:

The screenshot shows the MongoDB Aggregations interface. The aggregation pipeline is defined as follows:

```
1 {
2   directors: "John Landis",
3   "imdb.rating": { $gte: 7.5, $lte: 8 }
4 }
```

The output preview after the \$match stage (Sample of 4 documents) shows the following documents:

```
{
  "_id": "ObjectId('573a1397f29313caabce6d94')",
  "fullplot": "Faber College has one frat house so disreputable it will take anyone. ...",
  "imdb": {
    "year": 1978,
    "plot": "At a 1962 college, Dean Vernon Wormer is determined to expel the entire..."
  }
}
```

Other documents in the preview include:

```
{
  "_id": "ObjectId('573a1397f29313caabce76f7')",
  "plot": "Jake Blues, just out from prison, puts together his old band to save t..."
}
```

The interface includes buttons for "Generate aggregation", "Explain", "Run", "Options", "SAVE", "CREATE NEW", "EXPORT DATA", "EXPORT CODE", "PREVIEW", "STAGES", "TEXT", "WIZARD", and "Add stage".

Visualizamos resultado con Run:

The screenshot shows the MongoDB Aggregations interface after running the aggregation. The "Run" button is highlighted, and the "ALL RESULTS" section displays the following document:

```
{
  "_id": "ObjectId('573a1397f29313caabce6d94')",
  "fullplot": "Faber College has one frat house so disreputable it will take anyone. ...",
  "imdb": {
    "year": 1978
  }
}
```

The interface includes buttons for "Explain", "Run", "Options", "EXPORT DATA", "EXPORT CODE", and "Showing 1 - 4 count results".

Ejercicio 2

Muestra en un mapa la ubicación de todos los teatros de la base de datos.

Desde la colección 'theatres', seleccionamos la pestaña Schema, -> Analyze Schema y desde location, ampliamos y vemos campos 'address' y 'geo', donde visualizamos el mapa requerido::

localhost:27017 > CinemaDB > theaters

Documents 1.6K Aggregations Schema Indexes 1 Validation

Generate query + Reset Analyze Options

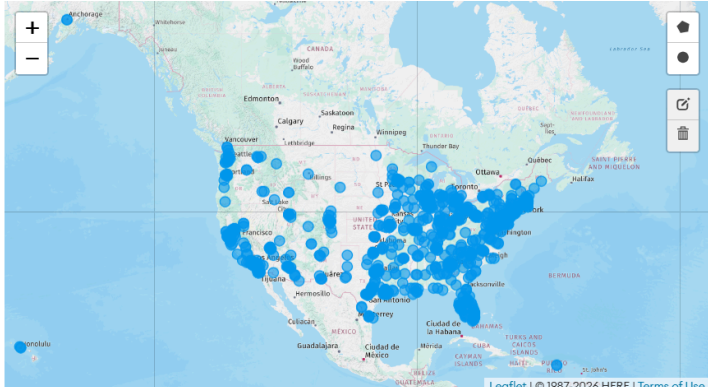
Project {}
Sort {}
Collation {}
Index Hint {}

Max Time MS 60000
Skip 0 Limit 0

EXPORT SCHEMA

This report is based on a sample of 1000 documents. [Learn more](#)

geo
coordinates



The map displays the geographical distribution of theaters across North America, including parts of Canada, the United States, and Mexico. Numerous blue dots represent individual theater locations, with a high concentration in the eastern United States and the Great Lakes region. The map includes a legend for 'geo' with a 'coordinates' field, and a scale bar at the bottom right indicating 20 miles.

fin sprint 5